Report 1 Preproccesing and aligment od the sequences

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Preproccesing and aligment of the sequences

Transciptomics

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1 Download

This script is used to download and preprocess the raw sequencing data from the RNA-seq experiments of the Ageing hallmarks exhibit organ-specific temporal signatures. Here, we focus on the male samples, particularly on the bone tissue at 1 and 24 age stages:

1.1 Download data

To access the data download, we use the program `fasterq-dump`. This is already installed in the Chaac server, so we can use subprocess from Python in order to make the download. To do so, we need the metadata for the experiment; this is stored in the GEO accession number GSE132040.

```
#Download the metadata table
wget https://ftp.ncbi.nlm.nih.gov/geo/series/GSE132nnn/GSE132040/suppl/ \
GSE132040%5FMACA%5FBulk%5Fmetadata.csv.gz -O data/Tabula_Muris_GSE_metadata.csv.gz

gunzip data/Tabula_Muris_GSE_metadata.csv.gz
```

```
The columns present in the metadata are:

- Sample name
- title
- source name
- organism
- characteristics: age
- characteristics: developmental stage
- characteristics: sex
- molecule
- description
- processed data file
- raw file
- BioSample
- Instrument Model
The organims from who the samples come form is ['Mus musculus C57/BL6']
The SRR id for the experiment C20_384Bulk_Plate1_S68 is SRR9126934
The SRR id for the experiment C20_384Bulk_Plate2_S68 is SRR9127010
The SRR id for the experiment D6_384Bulk_Plate2_S78 is SRR9127296
The SRR id for the experiment E3_384Bulk_Plate3_S99 is SRR9126859
The SRR id for the experiment F21_384Bulk_Plate1_S141 is SRR9127157
The SRR id for the experiment F7_384Bulk_Plate3_S127 is SRR9127369
The SRR id for the experiment J7_384Bulk_Plate1_S223 is SRR9127396
The SRR id for the experiment O9_384Bulk_Plate1_S345 is SRR9127293
So the SRR ids are:

- SRR9126934
- SRR9127010
- SRR9127296
- SRR9126859
- SRR9127157
- SRR9127369
- SRR9127396
- SRR9127293
```

With the IDs now we can download them

```
for srr in SRR_boneM:
    subP.run(["qsub", "-N", f"Download_{srr}", "src/jdl/Download.jdl", srr])
```

2 Preproscessing

The downloaded files are in `fastq` format. This is the general file format for short sequence data. The characteristics of this kind of file are that it not only contains the nucleotide sequence resulting from the sequencing reaction, but also has more properties such as the corresponding score for every nucleotide that is sequenced and the IDs for each sequence, having a total of 4 lines for each read sequenced. With this information provided by this format, we can deliver a FastQC (quality control) report with the objective of having the following statistics of our sequences:

- **Per Base sequence quality**
- **Per tile sequence quality**
- **Per sequence quality scores**
- **Per base sequence content**
- **Per sequence GC content**
- **Per base N content**
- **Sequence Duplication Levels**

This information in the reports is critical not only for knowing if the sequencing reaction procedure was done correctly, but also helps us to make decisions like trimming those bases that could have a low quality at the end or start of our sequences (as it frequently occurs with Illumina sequences).

```
#make the fastqc reports
qsub -N fastqc_report src/jdl/fastqc_rep.jdl data/
```

With the analysis product of the FastQC and MultiQC reports, we can make decisions about the preprocessing of the data:

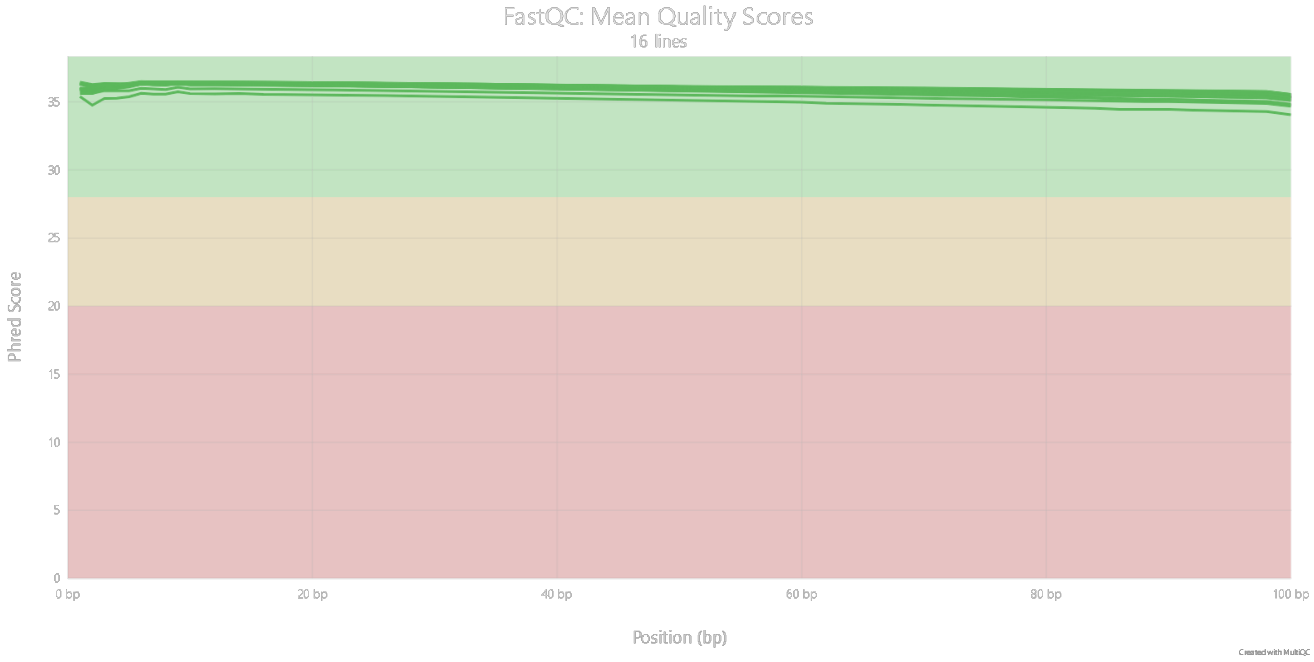


Figure 1: fastqc_per_base_sequence_quality_plot.png

After reviewing all the FastQC reports in the HTML view, we can see that the statistics related to the base quality and the sequence length are constant over all the samples and with good results, without having neither low scores nor inconstant sequence lengths. These results are characteristic of current NGS.

2.1 Possible problems

2.1.1 Base content

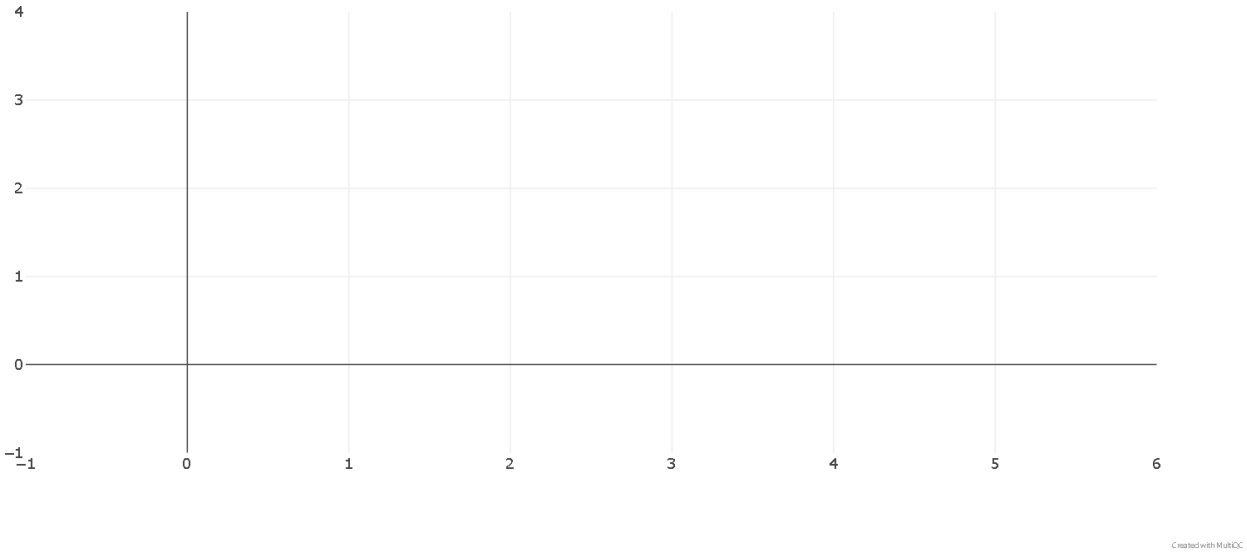


Figure 2: fastqc_per_base_sequence_content_plot.png

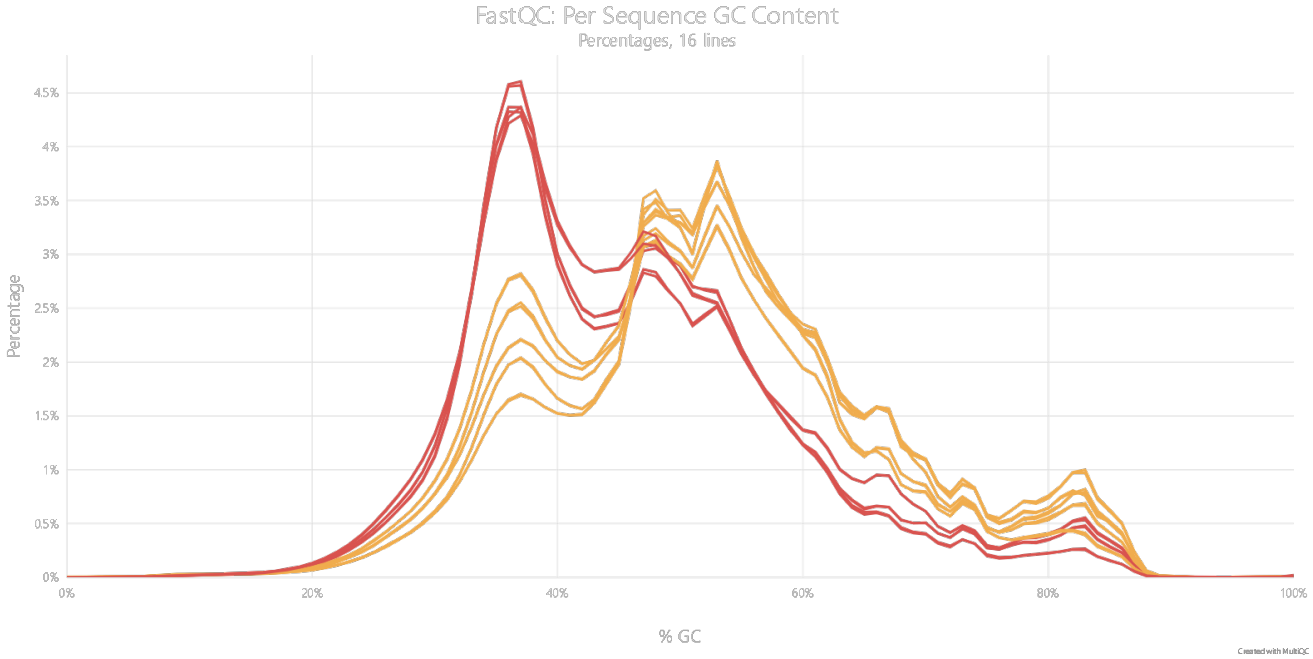


Figure 3: fastqc_per_sequence_gc_content_plot.png

One of the first problems we can see across all the samples is the base content related to the first bases in the reads. We can see that the base sequence content is very inconstant, having no proportional content of bases in a 1-15 range. As this problem is constant in all the libraries, it can be solved by trimming the sequences in their first base pairs.

```
#first iterarte over de data directory
for srr_dir in data/*/; do
    #extract the SRR id
    srr_id="${srr_dir%/*}"
    srr_id="${srr_id##*/}"
    echo "Procesing the $srr_id fastq files"
    #iterate over the files that are in the SRR directory
    for srr in "$srr_dir"*.fastq; do
        #now we check if it is the paired end file
        if [[ $srr == *_1.fastq* ]];then
            #we assume taht it has it paired file so we can obtain the id
            paired_1=$srr
            paired_2=${paired_1/_1.fastq/_2.fastq}
            echo "Files to be proccesed $paired_1 and $paired_2"
            qsub -N "$srr_id"_cleaning src/jdl/fastp_cleaning.jdl $paired_1 $paired_2
            break
        else
            #we are proccesing a unpaired-end file
            qsub -N "$srr_id"_cleaning src/jdl/fastp_cleaning.jdl $srr
        fi
    done
done
```

2.1.2 Duplication and overrepresented sequences

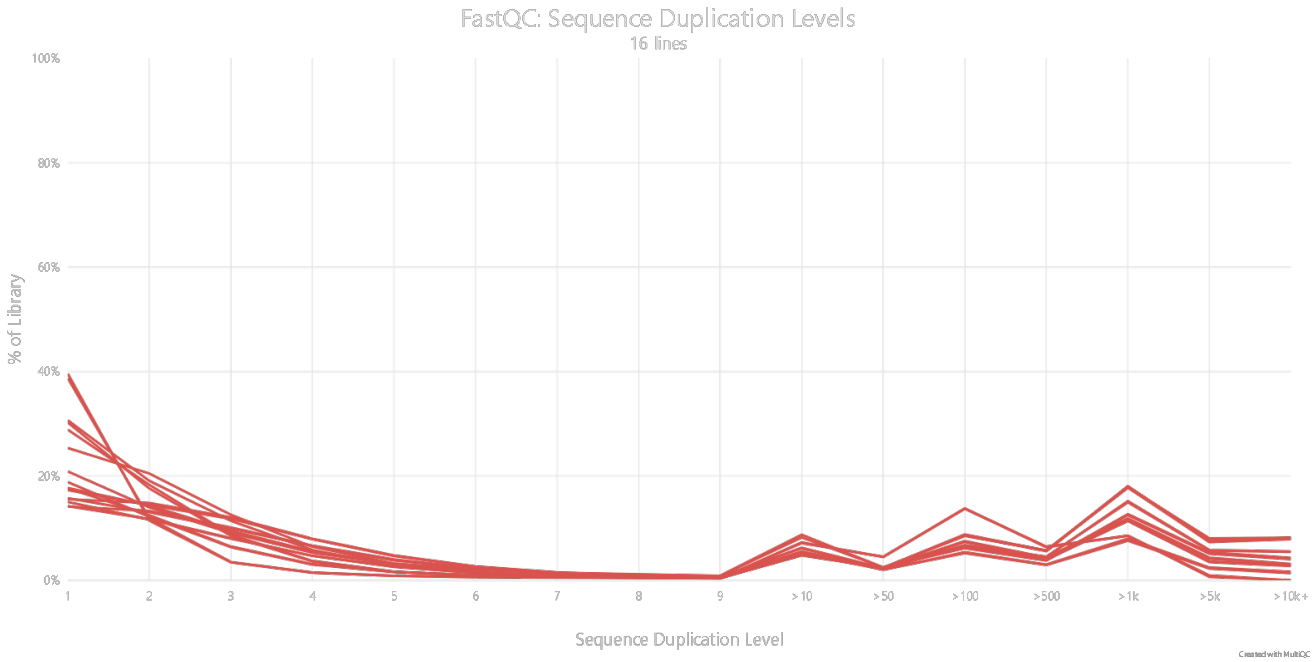
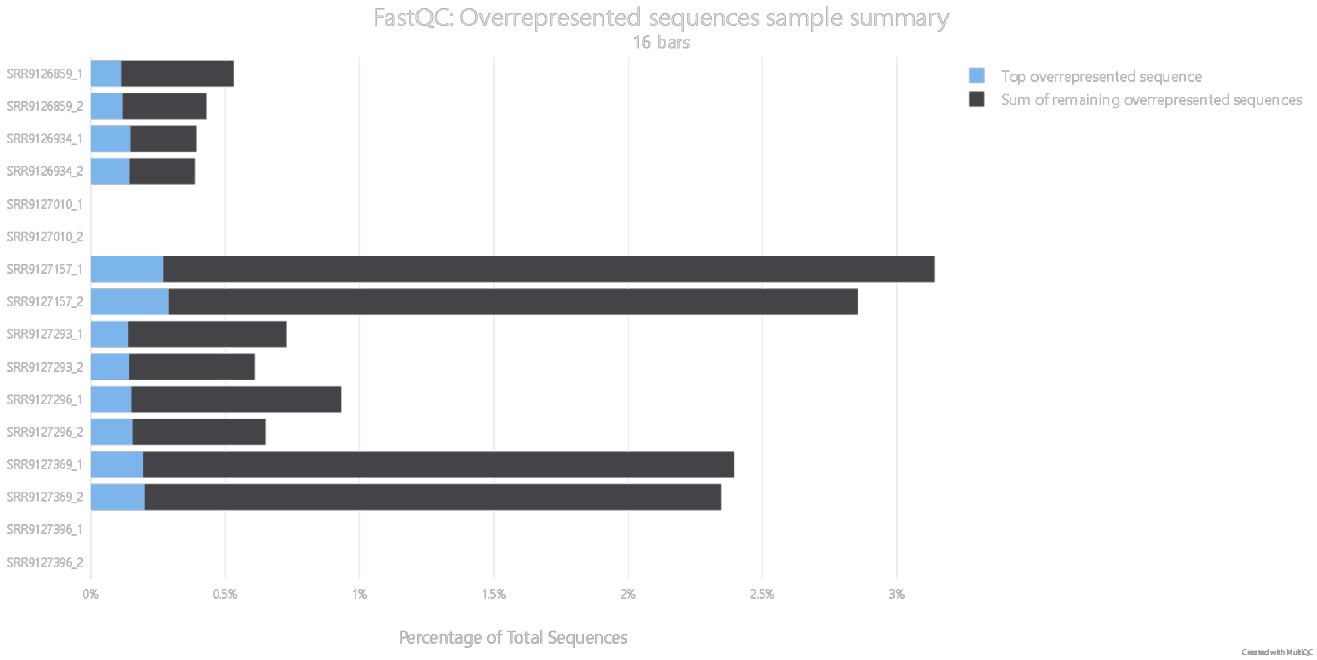
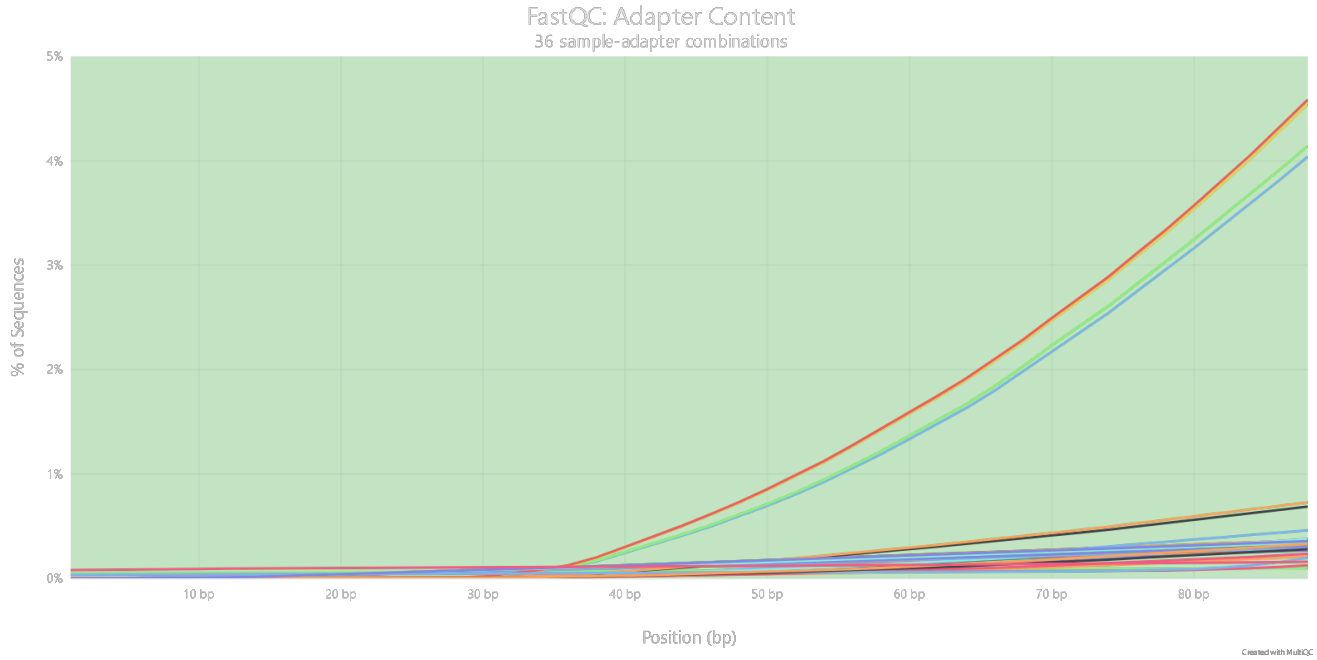


Figure 4: fastqc_sequence_duplication_levels_plot.png

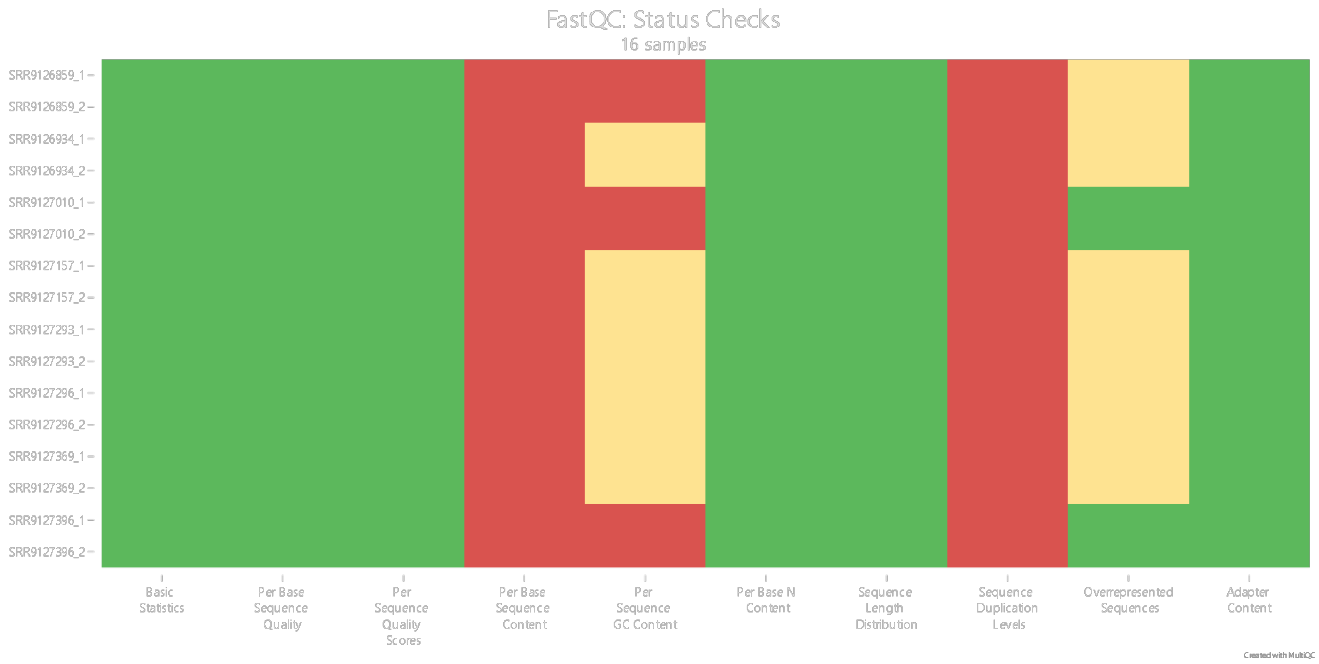


Another warning that all our samples have is the duplication level of sequences. We can see that this statistic can reach 72% of all the reads in some samples. Also, the majority of the SRR files have overrepresented sequences, with only those from the IDs SRR9127396 and SRR9127010 being the ones that do not have this problem.



In general, all the samples do not have a significant adapter content, with only two having a little representation of Nextera transposase sequence (SRR9126859 and SRR9127369). Despite this, it does not arise a warning.

2.2 General results



In this heatmap representing the problems across the SRR files, we can see that all the warnings are consistent across all the samples with few exceptions that we discussed above. So this gives us the first insight that our data is comparable, obviously with the precise normalization methods and preprocessing ones.

Now we make another time the QC report with the data processed by fastp.

```
#make the fastqc reports
qsub -N fastqc_report src/jdl/fastqc_rep.jdl data clean.fastq results/fastqc/fastp
```

We can see in the FastQC reports that the new trimmed sequences have improved at the start of all sequences after removing the first 15 base pairs of all our SRR files.

Once we have the files processed, we can start with the alignments of the data, so we finally save the relevant information such as the metadata table of our specific samples.

`The metadata tables looks:`

Table 1

src	name	months	srr_id	biosample	plate
C20_384Bulk_Plate1_S68	Bone_8	24	SRR9126934	SAMN11854715	Plate1
C20_384Bulk_Plate2_S68	Bone_16	1	SRR9127010	SAMN11854716	Plate2
D6_384Bulk_Plate2_S78	Bone_12	1	SRR9127296	SAMN11854792	Plate2
E3_384Bulk_Plate3_S99	Bone_21	24	SRR9126859	SAMN11854844	Plate3
F21_384Bulk_Plate1_S141	Bone_46	1	SRR9127157	SAMN11854896	Plate1
F7_384Bulk_Plate3_S127	Bone_32	1	SRR9127369	SAMN11854915	Plate3
J7_384Bulk_Plate1_S223	Bone_11	24	SRR9127396	SAMN11855147	Plate1
O9_384Bulk_Plate1_S345	Bone_35	24	SRR9127293	SAMN11855451	Plate1

Now we this pre-proccesed data and the metada data allready saved we can start with the next step, the same that is align the reads to a reference genome or in other words make all the reads sequenced to have sense in biological terms in the form that they have identification as places in the reference genome of the specie they came from.

3 Alignment

When you have the data clean and pre-proccesed we can asces the next step in the data processing, formaly we are talking about the alignment of the sequences to the reference genome, this references can be obtain by diferent data bases (like Genecode) and it can have annotatations of the genes, this is highly depended in the study coverage of the genome, for our fortune we are working with one of the animal models used to study human. The *mus musculus* genome index is the m36 realese available in genecode and indexed with the diferents programs that we are utilizing (hisat2 and star).

3.1 Comparison betwen aligners

Whenn you are doing a Diferential expression analysis starting with the raw sequence data a key step in the data processing is the alignment to the reference genome, this is curcial because is the step where the information stored in the sequence data takes a biological sense in the way that we can identified what genomic possition the read is refering to (in so many cases genes) and this identification is the milestone in the conslusion we can made in the following analysis.

For the above reasons it is important to compare diferent alignment tools not only for avoid the information lost also to make sure this information is sustented, it is known that there is not a perfect aligner and all alignments can not cover the 100% of the expression data, so the solution is to find the aligner that fits better to the experiemnt you are mading.

Here we analice the performance of two well characterized and widely used alignenment-tools in RNA-seq data `STAR` and `hisat2` in contrast with the use of paired and unpaired data.

For this we are going to use only the first SRR of the paired data of one alignment (in both tools) and then make the paired alignment also in both tools

When parsing hisat2 output:

- **Total aligned**: The percentage of all the reads aligned 'Overall alignment rate'
- **Non aligned**: the sum of the reads 'Aligned concordantly or discordantly 0 time' and 'Aligned 0 time'
- **Unique aligned**: the sum of the reads 'Aligned concordantly 1 time' and 'Aligned 1 time'
- **Multimaped**: The sum of the reads that were 'Aligned discordantly 1 time' and 'Aligned >1 times'
- **Time**: the total time from the table met report

For parsing the star output:

- **Total aligned**: The division bewtwen the sum of 'Uniquely mapped reads number' and 'Number of reads mapped to multiple loci' and 'Number of input reads'
- **Non aligned**: the sum of 'Number of reads unmapped: other', 'Number of reads unmapped: too short' and 'Number of reads unmapped: too many mismatches'
- **Unique aligned**: 'Uniquely mapped reads number'
- **Multimaped**: The sum of 'Number of reads mapped to multiple loci' and 'Number of reads mapped to too many loci'
- **Time**: The difference between 'Started mapping on' and 'Finished on'

With that information we can run the alignments while we parse the aligner results:

3.1.1 Clean data

```
# Create the scripts for the temporary paths and the scripts necessary
sp.run(["bash", "src/Generate_tempAlign.sh"])
# Iterate over all the SRR ids to run de corresponding alingments
for srr in SRR_ids:
    sp.run(["qsub", "src/jdl/Align_fatsq.jdl", "-S", srr, "-c"])
```

3.1.2 Raw data

```
sp.run(
    [
        "bash",
        "src/Generate_tempAlign.sh",
        "-t",
        "temp_align_raw/",
        "-r",
        "results/alignments_raw/",
    ]
)

for srr in SRR_ids:
    sp.run(
        [
            "qsub",
            "src/jdl/Align_fatsq.jdl",
            "-S",
            srr,
            "-t",
            "temp_align_raw/",
            "-r",
            "results/alignments_raw/",
            "-e",
            "align_errors_raw/",
        ]
    )
```

3.1.3 Merge the temporary results

```
#obtain the header
path_temp=temp_align_raw/parsed_results/
header=$(head -1 "$path_temp"$(ls "$path_temp" | head -1))

#put the header in the final file
echo "$header" > results/alignments_raw/alignments_parsed.csv
#for evey temporary table we read it and we append it to the final file
for temp_parse in "$path_temp"*.csv; do
    #append without the heeader (number of register 1)
    awk 'NR!=1' $temp_parse >> results/alignments_raw/alignments_parsed.csv
done
```

With this we obtain the folloeing tables with the performance of the aligners:

Table 2

SRR_id	Total_aligned	Non_aligned	Unique_aligned	Multimaped	Time(min)	Total_reads	Type	aligner
SRR9126859	0.832	0.399	0.454	0.093	29.58	37753296	PR	hisat
SRR9126859	0.781	0.219	0.644	0.137	72.58	23442600	PR	star
SRR9126859	0.834	0.166	0.673	0.161	29.58	23442600	UN	hisat
SRR9126859	0.910	0.090	0.707	0.203	27.15	23442600	UN	star
SRR9126934	0.827	0.405	0.491	0.064	29.58	49504973	PR	hisat
SRR9126934	0.785	0.215	0.694	0.091	58.50	30677615	PR	star
SRR9126934	0.830	0.170	0.684	0.146	29.58	30677615	UN	hisat
SRR9126934	0.911	0.089	0.738	0.173	25.32	30677615	UN	star
SRR9127010	0.843	0.382	0.493	0.073	29.58	51626561	PR	hisat
SRR9127010	0.821	0.179	0.698	0.123	96.30	32592759	PR	star
SRR9127010	0.845	0.155	0.688	0.157	29.58	32592759	UN	hisat
SRR9127010	0.921	0.079	0.731	0.190	33.23	32592759	UN	star
SRR9127157	0.830	0.404	0.480	0.077	29.58	36333435	PR	hisat
SRR9127157	0.767	0.233	0.682	0.086	39.20	21772221	PR	star
SRR9127157	0.833	0.167	0.670	0.163	29.58	21772221	UN	hisat
SRR9127157	0.918	0.082	0.726	0.192	15.67	21772221	UN	star
SRR9127293	0.865	0.350	0.546	0.063	29.58	47024412	PR	hisat
SRR9127293	0.826	0.174	0.739	0.088	54.82	30807452	PR	star
SRR9127293	0.871	0.129	0.733	0.138	29.58	30807452	UN	hisat
SRR9127293	0.941	0.059	0.776	0.165	20.27	30807452	UN	star
SRR9127296	0.837	0.390	0.509	0.063	29.58	8059574	PR	hisat
SRR9127296	0.786	0.214	0.712	0.074	8.62	5079340	PR	star
SRR9127296	0.841	0.159	0.707	0.134	29.58	5079340	UN	hisat
SRR9127296	0.919	0.081	0.764	0.155	3.35	5079340	UN	star
SRR9127369	0.781	0.464	0.416	0.083	29.58	38365169	PR	hisat
SRR9127369	0.694	0.306	0.614	0.080	37.70	22060603	PR	star
SRR9127369	0.784	0.216	0.636	0.148	29.58	22060603	UN	hisat
SRR9127369	0.879	0.121	0.703	0.176	15.73	22060603	UN	star
SRR9127396	0.882	0.319	0.569	0.061	29.58	48053937	PR	hisat
SRR9127396	0.864	0.136	0.756	0.109	82.48	32925807	PR	star
SRR9127396	0.895	0.105	0.753	0.142	29.58	32925807	UN	hisat
SRR9127396	0.943	0.057	0.781	0.162	30.32	32925807	UN	star

Table 3

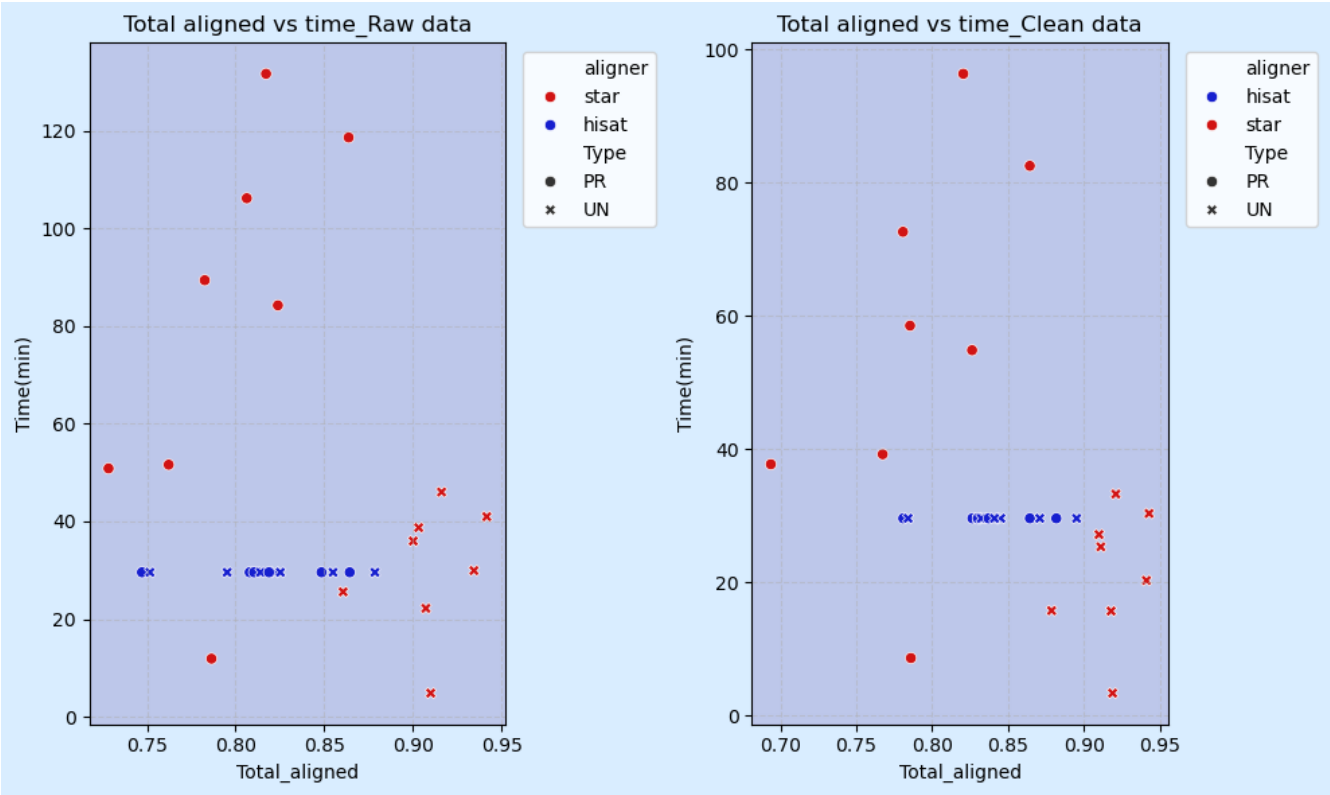
SRR_id	Total_aligned	Non_aligned	Unique_aligned	Multimaped	Time(min)	Total_reads	Type	aligner
SRR9126859	nan	nan	nan	nan	nan	nan	nan	nan
SRR9126859	0.806	0.194	0.683	0.123	106.23	23442600	PR	star
SRR9126859	0.795	0.205	0.664	0.131	29.58	23442600	UN	hisat
SRR9126859	0.900	0.100	0.713	0.187	35.98	23442600	UN	star
SRR9126934	0.808	0.430	0.478	0.057	29.58	50881029	PR	hisat
SRR9126934	0.783	0.217	0.701	0.081	89.43	30677615	PR	star
SRR9126934	0.812	0.188	0.688	0.124	29.58	30677615	UN	hisat
SRR9126934	0.903	0.097	0.744	0.159	38.73	30677615	UN	star
SRR9127010	nan	nan	nan	nan	nan	nan	nan	nan
SRR9127010	0.817	0.183	0.708	0.109	131.73	32592759	PR	star
SRR9127010	0.822	0.178	0.689	0.133	29.58	32592759	UN	hisat
SRR9127010	0.916	0.084	0.740	0.176	46.03	32592759	UN	star
SRR9127157	0.810	0.430	0.469	0.069	29.58	37411077	PR	hisat
SRR9127157	0.762	0.238	0.686	0.076	51.63	21772221	PR	star
SRR9127157	0.814	0.186	0.675	0.139	29.58	21772221	UN	hisat
SRR9127157	0.907	0.093	0.732	0.175	22.20	21772221	UN	star
SRR9127293	0.848	0.374	0.535	0.056	29.58	48245518	PR	hisat
SRR9127293	0.824	0.176	0.746	0.078	84.28	30807452	PR	star
SRR9127293	0.855	0.145	0.738	0.117	29.58	30807452	UN	hisat
SRR9127293	0.934	0.066	0.783	0.151	29.93	30807452	UN	star
SRR9127296	0.819	0.416	0.498	0.054	29.58	8269644	PR	hisat
SRR9127296	0.786	0.214	0.720	0.067	11.88	5079340	PR	star
SRR9127296	0.825	0.175	0.712	0.113	29.58	5079340	UN	hisat
SRR9127296	0.910	0.090	0.769	0.141	4.82	5079340	UN	star
SRR9127369	0.747	0.503	0.413	0.052	29.58	39210513	PR	hisat
SRR9127369	0.728	0.272	0.653	0.076	50.87	22060603	PR	star
SRR9127369	0.752	0.248	0.628	0.124	29.58	22060603	UN	hisat
SRR9127369	0.861	0.139	0.703	0.157	25.58	22060603	UN	star
SRR9127396	0.864	0.348	0.554	0.056	29.58	49420353	PR	hisat
SRR9127396	0.864	0.136	0.766	0.098	118.68	32925807	PR	star
SRR9127396	0.878	0.121	0.757	0.121	29.58	32925807	UN	hisat
SRR9127396	0.942	0.058	0.790	0.151	40.98	32925807	UN	star

3.2 Analysis

With the performance data provided by the aligners we can see the diferrences not only between aligners also between paired and unpaired data.

3.2.1 Total aligned vs time

The most common metric to evaluate the performance of an aligner is the total aligned reads, this is a key metric because it gives us an idea of how much information we are loosing in the alignment step, but also is important to evaluate the time that the aligner takes to do the alignment because this can be a bottleneck in the data processing.



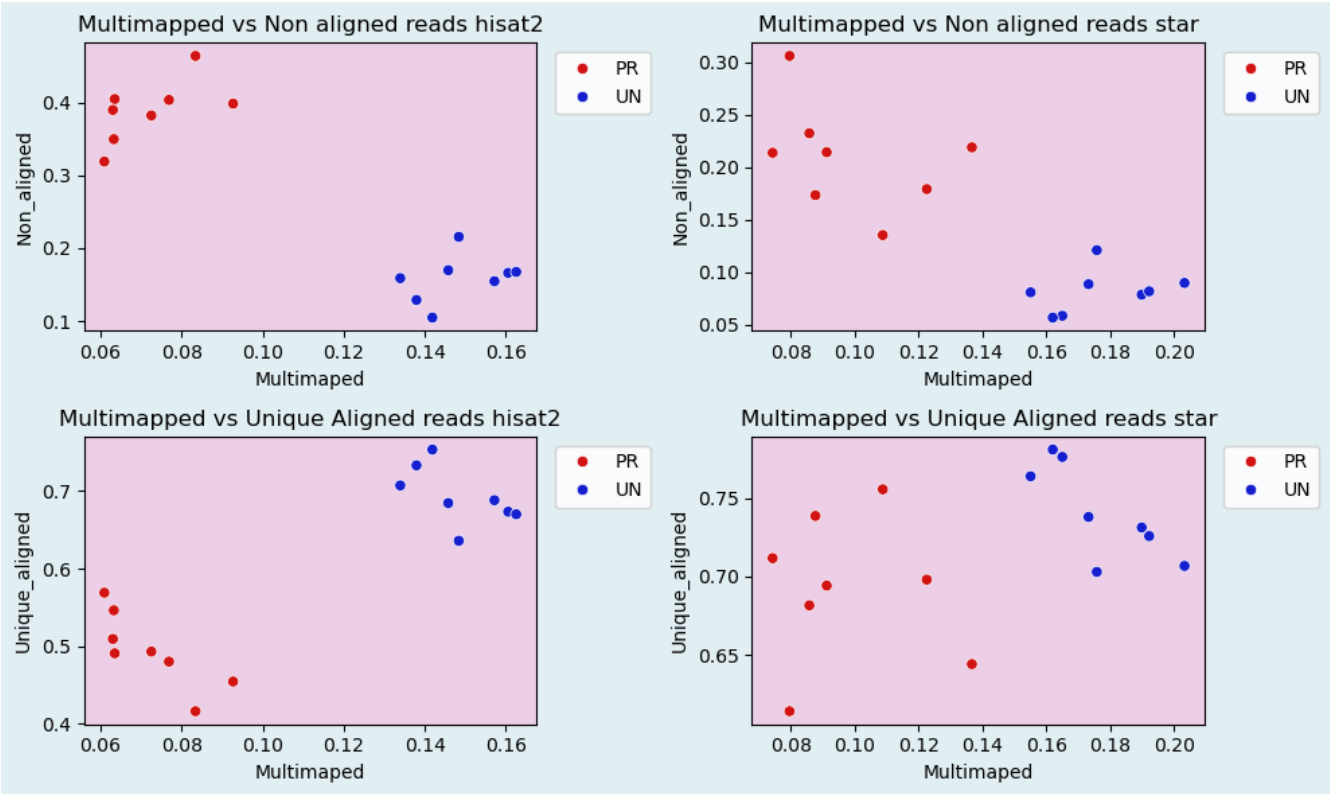
We can see that hisat has a constant time of performance, this does not tell us so much about the cost in aligning paired or unpaired data, in the other hand we can note that the cost in time for star has a important difference between paired and unpaired data, something interesting is that even the more time used and consequently the more computational resources are used the less total aligned reads are, this can have so many reasons, one is that there are so many multimapped reads that does not represent a trouble when aligning unpaired reads but are better discriminated when using paired data, so other matrics like the unique aligned reads or multimaped reads can give us more insights about the noise the unpaired data is adding to the alignment.

Other thing interesting in this case is that Star has a better performance using unpaired reads having more total aligned reads than hisat, even in less time, but whn using paired data the diferrence in total aligned reads does not seem to be big so the in terms of time hisat is better, but we have to see the other matrics to have a better idea of the performance of the aligners.

Finally the use of clean or raw data appears to not have a big impact in the total aligned reads, but it is important to note that the time used for the alignment is higher when using raw data, this can be due to the fact that the aligner has to deal with more noise in the data and this can make the alignment process more complex and consequently more time consuming, also some samples like `SRR9126859` and `SRR9127010` were imposible to map using hisat2 in raw data so we continue the further analysis with the clean-data alignments.

3.2.2 Multimapping

For seeing why this difference between paired and unpaired reads number of relative mapped reads we can view the non aligned and multimaped reads, this can give us an idea of how much noise the unpaired data is adding to the alignment and how much of this noise is being discriminated when using paired data, this is important beacuse this noise could be false positive mapped reads to those genomics locations.



As the scatter plots show the use of paired reads as it is suposed to be seems to have a "robust" alignment having more toruble in solving those sequences that are multimapped, this reflects the higher number of non aligned reads when using paired reads, it can be see as a more cofident alignment but comparing this multimaped sequences with the unique aligned reads we can see that even the use of unpaired reads has a high rate of unique mapped reads it also has a high rate of multimaped reads, this can be a problem because these multimaped reads can be false positives and can add noise to the data, this is something that is being discriminated when using paired reads, so the use of paired reads seems to be more accurate in the alignment process.

In this case we can see a clear trade off between the use of paired and unpaired reads, being the paired reads a more conservative approach but loosing some infomartion that the unpaired reads keep but with the risk of those multimapped reads that can be noise that the bivalent information of the paired reads can help to discriminate.

Finally we can see that star has a better performance than hisat2, this because it has more unique aligned reads and less multimaped reads, so even to the more strict aligment it recovers more unique aligned reads with similar levels of multimapped ones.

3.2.3 Conclusion

It is a fact that the time and resources invested in paired data aligment is higher than that for unpaired ones, so the question is clear, does it worth to invest this time and resources in the paired data aligment? Something clear is that this investment that in the most cases also involves more expensive experimental desings have a clear improve reducing multimapped reads that can be noise in the data, so the desision of using paired or unpaired data in the alignment step is a trade off between the time and resources invested and the accuracy of the alignment, so it is important to evaluate the specific needs of the experiment and the resources available before making a decision then the data also show us that the selection of the aligner coud be determinat in this king of trade offs and that is also experiment depedent desicion.

In thiscase we can see that two of the most used aligners does not have a determinat deiference in selecting one or another, but it can be viewed that star has a better performance in the paired data that could be in order of the time and reosurcers expended in those alignments, in the other hand we can se that although star has clearly a better total of alignment-time performance in unpaired data, the multimapping evidence can tell us that using hisat could be a better desicion with a not a big diference also with it constant time performance.

3.2.4 Code

All the code and extended results are disponible in Tabula_Muris_Transcriptomic_analysis git hub repository